

Telling Exponential Change from Linear Change

Answer Key

Algebra 1

Quick Answers

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|---|---|
| 1. (a) difference = 4, (b) ratio = 1.8, — | (d) $g(8) = 1280$ |
| 2. (a) ratio = 3, (b) first difference = 12, (c) second difference = 36 | 7. (a) plan A after 12 months = 980, (b) plan B after 12 months = 1006.10, — |
| 3. (a) $f(5) = 34$, (b) $g(5) = 128$ | 8. $x = 5$ |
| 4. (a) linear model after 4 years = 320, (b) exponential model after 4 years = 349.80 | 9. (a) straight-line value = 12000, (b) percentage-decay value = 12665.57 |
| 5. (a) ratio = 0.5, (b) the next value = 5 | 10. (a) $f(3) = 175$, (b) $g(3) = 172.80$, (c) $f(4) = 200$, (d) $g(4) = 207.36$, — |
| 6. (a) $f(4) = 130$, (b) $g(4) = 80$, (c) $f(8) = 210$, | |
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What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSF-LE.A.1 – *problems 1, 2, 3, 4, 5, 8*

HSF-LE.A.3 – *problems 6, 7, 9, 10*

Difficulty 1–5 of 5 across 27 tagged checks.

Common wrong answers

5. *If they got 40: continued the pattern by subtracting 10 each time, as a linear table would*

1. Table 5, 9, 13, 17. (a) A difference. (b) A ratio. (c) Which model, and why.

Solution (a): $9 - 5$.

Solution (b): $9 \div 5$.

The remaining differences are $13 - 9 = 4$ and $17 - 13 = 4$: constant. The remaining ratios are $13 \div 9 \approx 1.44$ and $17 \div 13 \approx 1.31$: not constant.

(c) — **instructor-judged.** A correct answer names the table linear and gives the constant-difference reason: every one-step increase in the input adds the same amount, 4,

4

1.8

to the output, while the ratios of successive outputs are not equal. Full credit for the label plus the constant-difference evidence; partial credit for the label alone.

2. Table 6, 18, 54, 162. (a) A ratio. (b) First difference. (c) Second difference.

Solution (a): $18 \div 6$.

3

Solution (b): $18 - 6$.

12

Solution (c): $54 - 18$.

36

The ratio stays at 3 ($54 \div 18 = 3$, $162 \div 54 = 3$) while the differences 12, 36, 108 keep growing — the signature of exponential growth, with rule $y = 6 \cdot 3^x$.

3. $f(x) = 4 + 6x$, $g(x) = 4 \cdot 2^x$. Find $f(5)$ and $g(5)$.

Solution (a): $4 + 6(5) = 4 + 30$.

$f(5) = 34$

Solution (b): $2^5 = 32$, so $4 \cdot 32$.

$g(5) = 128$

Both start at 4; five steps of +6 reach 34, five doublings reach 128.

4. Town of 200: model L adds 30 a year, model E multiplies by 1.15 a year. Populations after 4 years.

Solution (a): $200 + 30(4) = 200 + 120$.

320

Solution (b): $1.15^4 = 1.74900625$, so $200 \times 1.74900625 = 349.80125$, which rounds to two decimal places.

349.80

5. Table 80, 40, 20, 10. (a) The ratio. (b) The value at $x = 4$.

Solution (a): $40 \div 80$.

0.5

Solution (b): The pattern halves, so continue by halving rather than by subtracting: $80 \cdot \left(\frac{1}{2}\right)^4 = \frac{80}{16}$, which is $10 \div 2$.

5

6. $f(x) = 50 + 20x$, $g(x) = 5 \cdot 2^x$. Find $f(4)$, $g(4)$, $f(8)$, $g(8)$.

Solution (a): $50 + 20(4) = 50 + 80$.

$f(4) = 130$

Solution (b): $2^4 = 16$, so $5 \cdot 16$.

$g(4) = 80$

Solution (c): $50 + 20(8) = 50 + 160$.

$f(8) = 210$

Solution (d): $2^8 = 256$, so $5 \cdot 256$.

$g(8) = 1280$

At $x = 4$ the linear rule is ahead; by $x = 8$ the exponential rule is six times larger. An exponential eventually passes any linear rule.

7. Two \$500 plans: A adds \$40 a month, B multiplies by 1.06 a month. Values after 12 months, and which is better.

Solution (a): $500 + 40(12) = 500 + 480$.

980

Solution (b): $1.06^{12} = 2.01219647\dots$, so $500 \times 2.01219647 = 1006.0982\dots$, rounded to two decimal places. 1006.10

(c) — instructor-judged. A correct answer says plan B is worth more after twelve months and that its lead keeps widening, because a fixed percentage is applied to an ever-larger balance while the fixed monthly amount never changes. Full credit for naming plan B and giving the growing-base reason; partial credit for the comparison at twelve months alone.

8. $g(x) = 3 \cdot 2^x$. Solve $g(x) = 96$.

Solution: Divide both sides by 3: $2^x = 32$. Now write 32 as a power of 2: $32 = 2^5$, so the exponents match. Check: $3 \cdot 2^5 = 3 \cdot 32 = 96$ ✓ $x = 5$

9. Car worth \$24,000: straight line subtracts \$2400 a year, percentage method multiplies by 0.88 a year. Values after 5 years.

Solution (a): $24000 - 2400(5) = 24000 - 12000$. 12000

Solution (b): $0.88^5 = 0.5277319168$, so $24000 \times 0.5277319168 = 12665.5660\dots$, rounded to two decimal places. 12665.57

The percentage method leaves the car worth more here, because each year's loss is taken from a smaller value than the year before.

10. $f(x) = 100 + 25x$, $g(x) = 100 \cdot 1.2^x$. Values at $x = 3$ and $x = 4$, and where g overtakes.

Solution (a): $100 + 25(3) = 100 + 75$. $f(3) = 175$

Solution (b): $1.2^3 = 1.728$, so 100×1.728 . $g(3) = 172.80$

Solution (c): $100 + 25(4) = 100 + 100$. $f(4) = 200$

Solution (d): $1.2^4 = 2.0736$, so 100×2.0736 . $g(4) = 207.36$

(e) — instructor-judged. A correct answer states that the exponential model first overtakes the linear one at input 4, and justifies it from the two computed pairs: at input 3 the linear value 175 is still the larger, and at input 4 the exponential value 207.36 has passed 200. Full credit for the value 4 supported by both comparisons; partial credit for the value with only one comparison.